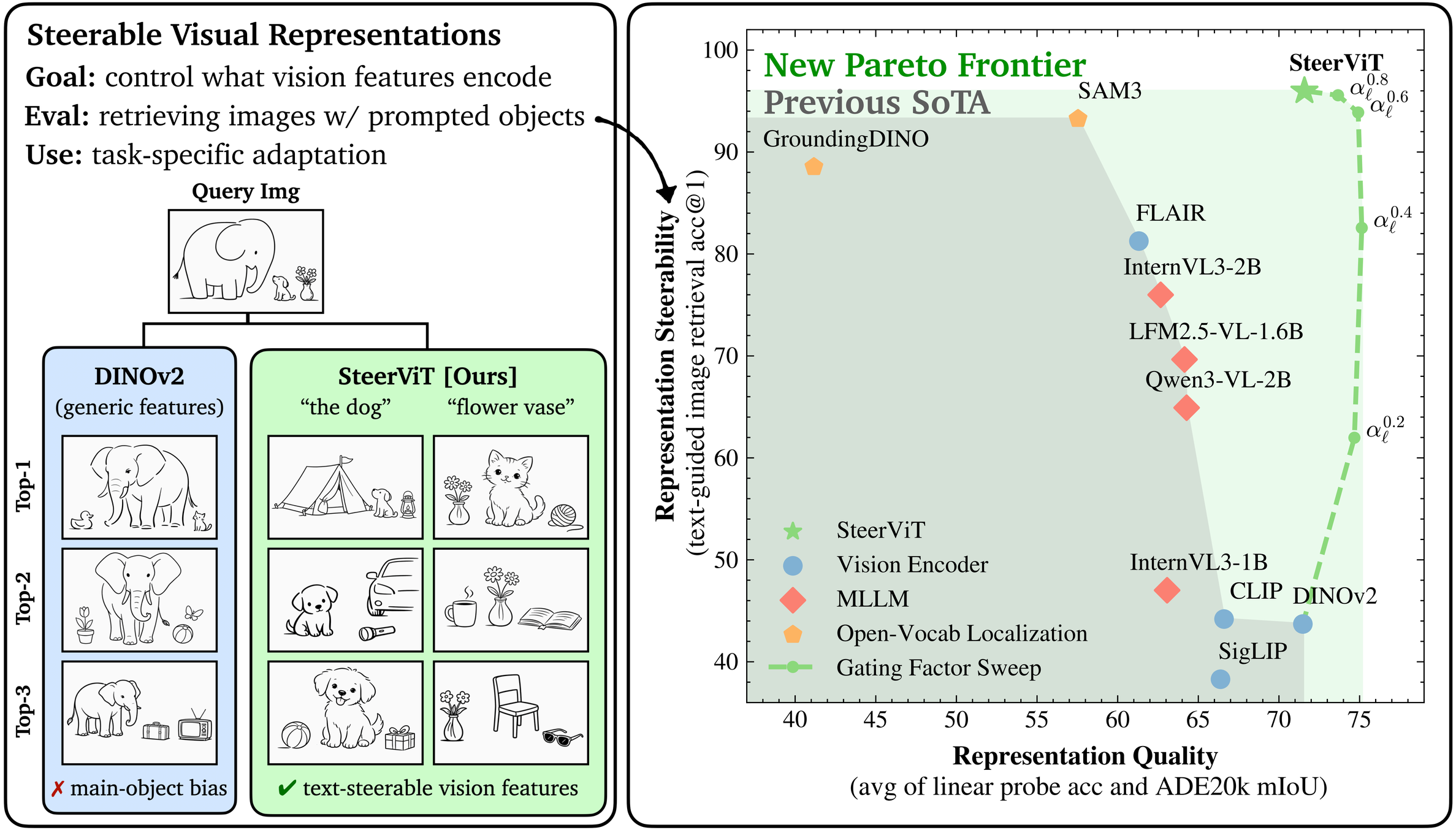
Steerable Visual Representations
Goal: control what vision features encode
Eval: retrieving images w/ prompted objects
Use: task-specific adaptation
Query Img
DINOv2
(generic features)
Top-1
Top-2
Top-3
✗ main-object bias
SteerViT [Ours]
"the dog"
"flower vase"
✔ text-steerable vision features
New Pareto Frontier
Previous SoTA
SteerViT
SAM3
GroundingDINO
FLAIR
InternVL3-2B
LFM2.5-VL-1.6B
Qwen3-VL-2B
InternVL3-1B
CLIP
DINOv2
SigLIP
$\alpha_\ell^{0.8}$
$\alpha_\ell^{0.6}$
$\alpha_\ell^{0.4}$
$\alpha_\ell^{0.2}$
SteerViT
Vision Encoder
MLLM
Open-Vocab Localization
Gating Factor Sweep
Representation Steerability
(text-guided image retrieval acc@1)
Representation Quality
(avg of linear probe acc and ADE20k mIoU)
100
90
80
70
60
50
40
40
45
50
55
60
65
70
75